

**Multitasking and Risky Decisions: A Study exploring the effects of
Multitasking on Risky Decisions**

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Abstract

(Written by X. Franco; edited by X.Franco)

Studies have explored the effects of multitasking on decision making, primarily focusing on the switch cost effect. However, the impact of metacognitive load on multitasking, particularly in situations involving risky decision-making and high reward, has yet to be thoroughly investigated. This study hypothesizes that multitasking will increase the probability of making risky decisions, as a participant's divided attention may lead them to rely on basic best outcome ideals and therefore favor high-risk, high-reward options. In addition, it is hypothesized that as the reward for the risky choice increases, the likelihood of selecting that option will increase for both the single-task and multitasking conditions. In the single task condition, participants chose between option A or B, each associated with a different likelihood of obtaining a certain number of points. In the multitasking condition, participants completed an additional task with consequences while also choosing between A and B. Participants were randomly assigned to first complete either the multitasking condition or the single task condition. It was expected that in the multitasking condition, participants would evaluate the tradeoff between reliability and potential reward, thus demonstrating greater risk-taking compared to the single task condition. Our results, however, showed no significant differences between conditions, suggesting that factors outside of the metacognitive load may influence risky decision making and that further research should be conducted to better understand the neural mechanisms involved in the processes of decision making, particularly those based on value-based decisions.

Introduction

(Written by X. Franco & M. Merkel; edited by X. Franco & M. Merkel)

Decision-making is a widely studied concept with large amounts of research examining its neural circuits and cognitive processes. Within this domain is a subcategory of interest, this being the effect that imposing factors have on decision-making. In this proposed experiment, we will explore the effects of multitasking on decision-making, which places more demand on cognitive processes. These processes involve frameworks such as neural networks in the ventromedial prefrontal cortex (VmPFC), which play a role in value-based decision making (Fellows & Farah). These values are weighed against computational frameworks such as the neural systems of rewards and risks to predict outcomes. We aim to test these value-based decisions and how they coincide with high-risk, high-reward outcomes in tangent with the metacognitive load of multitasking.

In past research, the effects of multitasking have primarily been concerned with the switch cost that multitasking has on behavior, as the demands it imposes on the brain can lead to more frequent errors within a task. Studies such as Madore, K. P., & Wagner, A. D. (2019) and Rogers, D. R., & Monsell, S. find that as the mental load increases due to multitasking, it could affect decision-making.

Our proposed study builds on previous research by exploring the effects of multitasking outside of solely the switch cost. Switch cost refers to the consequences that appear from switching from one task to another. We will look beyond the behavioral aspects, specifically with concerns towards how the cognitive load of multitasking will increase the probability of risky decisions to favor high-risk, high-reward options. Thus, we aim to explore *how does multitasking affect the probability of making risky decisions?*

To address this question, participants will be randomly assigned to complete both the single-task risk and the multi-task risk condition. These tasks will test the probability of risky decision-making when multitasking is present compared to when it is not. Included in the multitasking factors will be rapid decision-making. Our experiment will also consist of a leaderboard to persuade our participants to continuously make decisions they believe will push them ahead. Our dependent variables will consist of

how often participants make risky decisions based on the two given tasks (multitasking versus single-task), seeking risk to acquire the reward of being on the leader board. This experiment will work to highlight that when faced with multitasking tasks, participants' divided attention will lean on outcome ideals, leading them to choose high-risk, high-reward options more often.

We hypothesized that multitasking would increase the probability of making risky decisions, as dividing attention reduces executive function, which therefore limits participants' ability to carefully evaluate all outcomes (which decreases the cognitive resources available to evaluate options/outcomes). As a result, participants may be more likely to use simpler, more automatic decision-making processes, which would favor the high-risk, high-reward option. This can be explained by prior research that suggests that an increase in cognitive load will diminish the deliberation process and push participants to more heuristic choices (Shiv & Fedorikhin).

In addition, we hypothesized that as the magnitude of the reward for the risky choice increases, the probability of choosing the risky option will exponentially increase for both single-task and multi-task risk tests. This is based on prior research of the Prospect Theory, which explains that higher rewards increase the perceived value and attractiveness of the risky option, which makes it more likely to be selected even when there is greater uncertainty (Kahneman & Tversky).

Methods

(Written by J. Jiang; edited by J. Jiang & M.Merkel)

Participants

There are a total of 37 participants completed the study. Participants were recruited from a college student population and were between 18 and 25 years old. Participation was voluntary. To protect anonymity, the experiment did not collect names or identifying information. Each participant was assigned an anonymous participant ID through the experiment's backend system.

Design

This study used a within-subjects design to examine whether multitasking affects risky decision-making. Each participant completed two task conditions: a single-task

(ST) condition and a multi-task (MT) condition. In the ST condition, participants completed only a risky decision-making task. In the MT condition, participants completed the same risky decision-making task while also performing a concurrent digit-monitoring task. To reduce order effects, participants were randomly assigned to complete either the single-task condition first or the multi-task condition first.

The main dependent variable was participants' probability of choosing the safe option, Option A. Choosing Option A was interpreted as a safer decision, while choosing Option B was interpreted as a riskier decision.

The experiment was built as a web-based task using HTML and JavaScript. Trial data were automatically stored using a Google Apps Script backend connected to Google Sheets. The program assigned anonymous participant IDs, randomized task order, generated trial-specific reward values, calculated outcomes, updated participants' total scores, and saved trial-level data.

Stimuli and Task Structure

On each trial, participants chose between two options displayed on the left and right sides of the screen. Option A on the left was the safe option and always provided an 80% chance of earning 10 points. Option B on the right was the risky option and provided a 20% chance of earning X points. The value of X was randomly generated on each trial and ranged from 1 to 60 points.

During the main trials, a live leaderboard was displayed at the bottom of the screen. The leaderboard showed the top five scores from previous participants and the current participant's running total score. The purpose of the leaderboard was to increase motivation and make total points more salient during decision-making.

In the MT condition, participants completed the same choice task while also monitoring a stream of randomly presented digits in the center of the screen. Digits appeared one at a time, changing every 1.5 seconds. Participants were instructed to press the space bar whenever the number 3 appeared. If participants pressed the space bar for an incorrect number or missed the number 3, they received a 5-point penalty. This secondary task was included to increase cognitive load and divide attention during risky decision-making.

Procedure

Participants first saw a welcome screen and entered a participant ID. The system then assigned them an anonymous backend ID and randomly determined whether they would complete the ST or MT condition first. Before each main task condition, participants completed three practice trials to become familiar with the task. Practice trials were not included in the final dataset.

In the single-task condition, participants completed 20 main trials. On each trial, they selected either Option A or Option B by clicking the corresponding side of the screen. After each choice, the outcome was shown briefly as feedback, such as “+10” or “+0,” and the participant’s total score was updated.

In the multi-task condition, participants also completed 20 main trials. The risky decision task was identical to the ST condition, but participants simultaneously monitored the digit stream and pressed the space bar whenever the digit 3 appeared. Penalties from the digit-monitoring task were subtracted from the participant’s total score. After completing both task conditions, participants saw a final leaderboard and a completion message.

Data Collection and Analysis Pipeline

The Google Sheets backend recorded trial-level data for each participant, including participant ID, task condition, trial number, choice, X value, points earned, and task order. Practice trials were excluded from analysis. For each participant and condition, the probability of choosing Option A was calculated by dividing the number of Option A choices by the total number of trials in that condition. Because Option A was the safer option, lower values of this measure indicated greater risk-taking.

The main analysis compared the probability of choosing Option A between the ST and MT conditions. A paired-samples comparison was used because each participant completed both conditions. Task order was recorded so that ST-first and MT-first participants could be compared to check whether completing one condition first influenced later choices.

Results & Discussion

(Results written and edited by S. Cantu Vazquez; Discussion written and edited by M.Merkel)

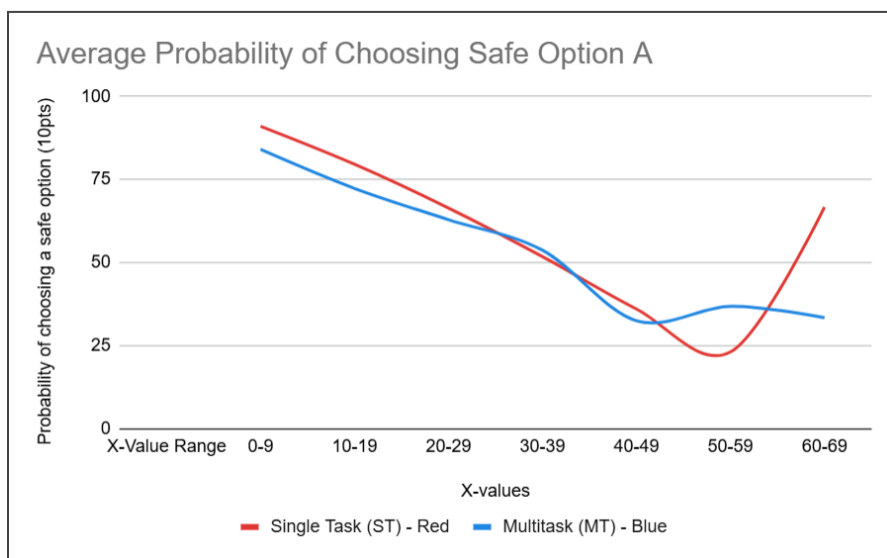
Results

Before, we hypothesized that those in the multitasking group would be more prone to making riskier decisions. Meaning, they (multitask group) would choose Option B more compared to those in the single-task session. Our current findings (data) do not support the above hypothesis. First, by viewing both figures (Fig 1a and 1b) below, we can see a consistent trend in both the Single and multitasking sessions, where they appear very similar in choosing a more risky option (option B). In addition, when viewing for a significant difference, the P-value ended up being 0.7, indicating there was no significant difference between the single versus the multitasking session in showing which session was more prone to choosing a riskier option. For our 2nd hypothesis, we predicted that as the X-value (magnitude of reward) went up, risky choices would also increase. By viewing both Fig 1a and 1b, we can see a consistent trend with this concept. In both the single and multitask sessions, as X-value increases, the probability of choosing a safe option (Option A) diminishes steadily. When viewing for a significant difference on this concept, the P-value ended up being 0.01. From this result, our data does support our 2nd hypothesis, indicating that as the X-value increases, so does the probability of choosing a riskier option.

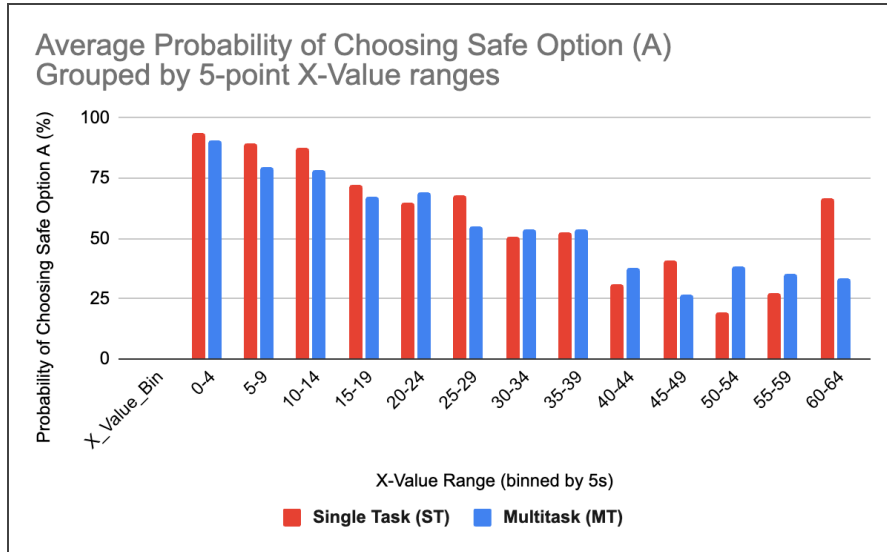
Figure 1a and 1b

Average Probability of Choosing the Safe Option by X-Value Range

a



b



Discussion

Contrary to our hypothesis that multitasking increases the probability of making risky decisions, our results had no significance. This may be because multitasking reduces cognitive resources which, in turn, can lead to less confidence when evaluating the risky-decision and, therefore, choosing the safe/default option. This can be supported by prior research that suggests that an increase of cognitive load will diminish the deliberation process and push participants to more heuristic choices (Shiv & Fedorikhin). In this context, heuristic decision making may not favor the higher risk, greater reward option, but instead promote the safer, more consistent option. In addition, the penalty of losing 5 points when pressing the space bar for an incorrect number or missing the number 3 may have furthered the conservative strategy, as not only did multitasking divide the participants attention but the penalty added another layer of risk. Finally, the aspect of competition involving the leader board could add another reason for the null hypothesis as it emphasizes the possibility of falling behind when choosing the risky option.

Consistent with our secondary hypothesis, participants were more likely to choose the risky-option as the magnitude of the reward increased, and this was reflected in the significant findings. This can be explained by Prospect Theory which suggests that higher rewards increase the perceived value and attractiveness of risky

options, which makes them more likely to be selected even when there is greater uncertainty. This was observed within our study as the probability of the participants choosing the risky option proportionally increased with the magnitude of the reward.

Further examination of the data, however, revealed a deviation from the overall trend of choosing a more risky option for a higher reward. For the X-values of 60-69 in the single task portion, there was an unexpected increase of choosing the safe option. One possible cause of this is that, at these reward levels, the participants became more sensitive to the possibility of receiving no reward from choosing the risky option, so they reverted to the safe option. This can be explained by the other component of Prospect Theory in which large potential losses are weighted more heavily over equivalent gains (Kahneman & Tversky). Therefore, in this range of rewards, a pull is observed between reward attraction and loss-aversion as rewards increase.

Several limitations should be acknowledged. First, our study was restricted to college students, and as a result, generalizability is restricted. A demographic that has a wider age range and includes those with different levels of education may change our results. Furthermore, our sample size had only 37 participants, which reduces the statistical power. A larger sample size could help decrease the p-value and lead to a significant result. Lastly, this experiment may not replicate real-world decision making processes, as situations that involve risk and reward are often more complex and have more emotional input.

Neuroscience Extensions

(Written and edited by Samuel Cantu Vazquez)

Furthermore, to extend our research to broaden our understanding of risky decision-making. Our current study can test participants with lesions (damage) to the Ventromedial Prefrontal Cortex (VPC), and compare results with healthy subjects on our previous tests (single versus multi). Currently, it is thought that the VPC has an influential role in decision-making (Masama et al., 2022). Since there is an association between the VPC and decision making, we could see how each group performs

differently (healthy versus lesions). A hypothesis could be as follows: for those who have lesions in the VPC, their performance should show poorer (riskier) results when compared to healthy individuals. Since proper judgment for those with lesions in the VPC may be hindered, a possible theory suggests that their decisions may lean more towards risky decisions, especially during the multitasking session. Our findings to view this outcome will strictly come from data that is collected anonymously from both groups (lesions versus healthy individuals). Studying this particular field may enhance our current understanding as to which parts of the brain could be linked to having an impact on our judgment in our day-to-day lives. Issues with conducting an experiment is finding sufficient subjects who have lesions in their VPC. Other issues such as how much damage to the VPC must be also be taken into account.

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Contributor Book

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Abstract: Ximena Franco

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